Chapter 5. Functions

Most of you are familiar with the concept of a *function*, such as $y = x^2$ which gives you a value of y for any given value of x. Such an idea can be extended to more general situations in which x and y are elements of certain sets (not necessarily real numbers) and the correspondence need not be given by a formula.

5.1. The Definition of Function

Definition 5.1 Let A and B be sets. A *function* or a *mapping* from A to B, written $f : A \longrightarrow B$, is a relation from A to B such that for every element a in A, there exists a unique $b \in B$ such that $(a, b) \in f$.

Since f is a relation, the set A is called the *domain* of f, denoted by dom f. The set B is called the *codomain* of f.

Remark The element b is unique for any given element $a \in A$, that is, for any $a \in A$ and for any $b, c \in B$, if $(a, b) \in f$ and $(a, c) \in f$, then we must have $b = c$. We write $b = f(a)$ and call it the *image of a under f*. We also refer to a as the *pre-image of b under f* and we say that f *maps a into b*.

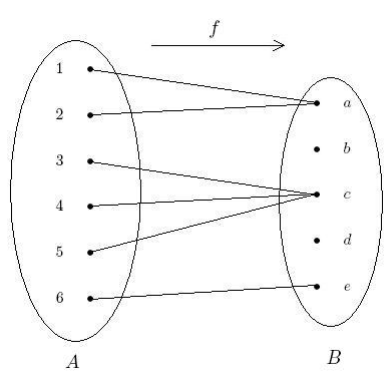
Definition 5.2 The *range* of f, denoted by ran f, is defined by

$$\text{ran } f = \{b \in B : b = f(x) \text{ for some } x \in A\} = \{f(x) : x \in A\} \subseteq B.$$

Example 5.3 Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{a, b, c, d, e\}$. Then

$$f = \{(1, a), (2, a), (3, c), (4, c), (5, c), (6, e)\}$$

defines a function:



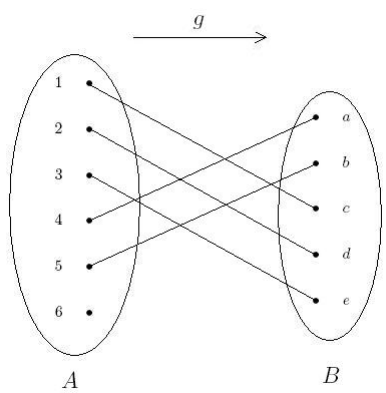
In this example, a is the image of both 1 and 2 under f, that is, $f(1) = f(2) = a$. In other words, 1 and 2 are pre-images of a. Similarly, we have $f(3) = f(4) = f(5) = c$ and $f(6) = e$. Note that b and d are not the images of any element in A, that is, they have no pre-images in A.

The set A is the domain of f and B is the codomain of f. The range of f is given by

$$\text{ran}\, f \;=\; \{a, c, e\} \subseteq B. \qquad \square$$

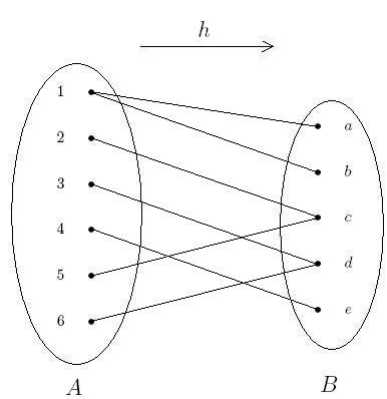
Example 5.4 Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{a, b, c, d, e\}$. The following is NOT a function from A to B:

$$g = \{(1, c), (2, d), (3, e), (4, a), (5, b)\}$$



Example 5.5 Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{a, b, c, d, e\}$. The following is again NOT a function from A to B:

$$h = \{(1, a), (1, b), (2, c), (3, d), (4, e), (5, c), (6, d)\}$$



Example 5.6 Let A and B be sets with $b \in B$ a fixed element. Then the function $f : A \longrightarrow B$ defined by $f(a) = b$ for all $a \in A$ is called the *constant function* from A to B. Note that b is the image of all the elements in A, that is, every element in A is the pre-image of b. The range of this constant function f is given by $\operatorname{ran} f = \{b\}$. □

Example 5.7 Let A be a set. Then the function $id_A : A \longrightarrow A$ defined by $id_A(a) = a$ for all $a \in A$ is called the *identity function* of A. Every $a \in A$ is the image of itself. The domain, codomain and range of id_A are all equal to A. □

We finish this section by defining the equality of two functions:

Definition 5.8 Let A and B be sets, and let

$$f : A \longrightarrow B \qquad \text{and} \qquad g : A \longrightarrow B$$

be two functions from A to B. Then f and g are said to be *equal*, written $f = g$, if $f(a) = g(a)$ for all $a \in A$.

5.2. Injective, Surjective and Bijective Functions

Injective Functions

Definition 5.9 A function $f : A \longrightarrow B$ is called *injective* or *one-to-one* if

$$\forall x, y \in A,\ f(x) = f(y) \implies x = y,$$

or equivalently,

$$\forall x, y \in A,\ x \neq y \implies f(x) \neq f(y).$$

In other words, an injective function f maps distinct elements of A into distinct elements of B. However, if there exist distinct elements x and y in A such that $f(x) = f(y)$, then f is not injective. For instance, the function in Example 5.3 is not injective since $f(1) = f(2) = a$.

To show that a function is injective, we usually use a direct proof: for any $x, y \in A$, assume $f(x) = f(y)$ and proceed to prove that $x = y$.

Example 5.10 Let the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ be defined by $f(x) = 5x - 1$. Show that f is injective.

Proof. For any $x, y \in \mathbb{R}$, assume that $f(x) = f(y)$. Then we have

$$5x - 1 = 5y - 1 \implies 5x = 5y \implies x = y.$$

Therefore f is injective. □

Example 5.11 Let the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ be defined by $f(x) = x^2 + 1$. Determine whether f is injective.

Solution. Since $f(1) = 2 = f(-1)$, it follows that f is not injective. □

Example 5.12 Let the function $f : [0, \infty) \longrightarrow \mathbb{R}$ be defined by $f(x) = x^2+1$. Show that f is injective.

Proof. For any $x, y \in [0, \infty)$, assume that $f(x) = f(y)$. Then we have

$$x^2 + 1 = y^2 + 1 \implies x^2 = y^2 \implies x^2 - y^2 = 0 \implies (x - y)(x + y) = 0.$$

Thus $f(x) = f(y)$ implies either $x = y$ or $x = -y$. The latter case is only possible when $y = 0$ (but then we have $x = y = 0$, which is the first case); otherwise it is not possible since $x = -y$ would imply that x is negative. Therefore we must have the first case, and we are done. □

Remark Example 5.11 and 5.12 suggest that the <u>domain</u> of a function is important when we determine whether the function is <u>injective</u> or not. We should not just look at the formula of f.

Surjective Functions

Definition 5.13 A function $f : A \longrightarrow B$ is called *surjective* or *onto* if

$$\forall y \in B,\ \exists x \in A \text{ such that } f(x) = y.$$

In other words, if every element of the codomain B is the image of some element of A, that is, <u>ran $f = B$,</u> then f is surjective.

Example 5.14 Let the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ be defined by $f(x) = 5x - 1$. Show that f is surjective.

Proof. We have to prove that for any $y \in \mathbb{R}$, there exists $x \in \mathbb{R}$ such that $f(x) = y$. We are trying to solve an equation where y is given. For any $y \in \mathbb{R}$, if the equation $y = 5x - 1$ has a solution x in $\mathbb{R}$, then the function f is surjective. But it is clear in this example that we can rearrange the equation so that x is the subject of the equation:

$$x = \frac{y+1}{5} \in \mathbb{R}.$$

To summarize, we let $y \in \mathbb{R}$ and choose $x = (y + 1)/5$. Then $x \in \mathbb{R}$ and

$$f(x) = f\left(\frac{y+1}{5}\right) = 5\left(\frac{y+1}{5}\right) - 1 = (y + 1) - 1 = y.$$ □

To show that a function $f : A \longrightarrow B$ is not surjective, we need to prove that there exists an element $y \in B$ such that $y \neq f(x)$ for any $x \in A$. For instance, the function in Example 5.3 is not surjective since we have b (and d) in B such that it is not the image of any element in A.

Example 5.15 Prove that the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = x^2 + 1$ is not surjective.

Proof. Take $y = 0$. Then there is no $x \in \mathbb{R}$ such that

$$0 = f(x) = x^2 + 1,$$

as $x^2 = -1$ has no real solution. Therefore f is not surjective. □

Example 5.16 Prove that the function $f : \mathbb{R} \longrightarrow [1, \infty)$ defined by $f(x) = x^2 + 1$ is surjective.

Proof. Given any $y \geq 1$, we would like to find an x such that $y = f(x) = x^2 + 1$. Since $y \geq 1$, it is possible to take $x = \sqrt{y-1}$. Thus for any $y \geq 1$, we choose $x = \sqrt{y-1}$. Then $x \in \mathbb{R}$ and

$$f(x) = f(\sqrt{y-1}) = (\sqrt{y-1})^2 + 1 = (y-1) + 1 = y.$$ □

Remark Example 5.15 and 5.16 suggest that the <u>codomain</u> of a function is important when we determine whether the function is <u>surjective</u> or not. Again, we should not just look at the formula of f.

Bijective Functions

Definition 5.17 A function $f : A \longrightarrow B$ is called *bijective* or a *one-to-one correspondence* if it is both injective and surjective.

In other words, f is bijective if <u>for any</u> $y \in B$, there exists a <u>unique</u> $x \in A$ such that $f(x) = y$.

Example 5.18 From Example 5.10 and 5.14, the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ given by $f(x) = 5x - 1$ is bijective. □

Example 5.19 From Example 5.12 and 5.16, the function $f : [0, \infty) \longrightarrow [1, \infty)$ given by $f(x) = x^2+1$ is bijective. □

Example 5.20 Prove that the function $f : \mathbb{R} \setminus \{2\} \longrightarrow \mathbb{R} \setminus \{5\}$ defined by $f(x) = \dfrac{5x+1}{x-2}$ is bijective.

Proof. Injective: For any $x, y \in \mathbb{R} \setminus \{2\}$, assume $f(x) = f(y)$. Then we have

$$\frac{5x+1}{x-2} = \frac{5y+1}{y-2} \implies (5x+1)(y-2) = (5y+1)(x-2) \implies -10x + y = -10y + x \implies x = y.$$

Thus f is injective.

Surjective: Given any $y \in \mathbb{R} \setminus \{5\}$, we try to find an $x \in \mathbb{R} \setminus \{2\}$ such that $y = f(x) = \frac{5x+1}{x-2}$. Choose $x = \frac{2y+1}{y-5}$. Note that this x is well-defined as $y \neq 5$. Note also that this x must not equal to 2, for if $x = \frac{2y+1}{y-5} = 2$, then we have $2y + 1 = 2y - 10$, which is impossible. Now

$$f(x) = f\left(\frac{2y+1}{y-5}\right) = \frac{5\left(\frac{2y+1}{y-5}\right) + 1}{\frac{2y+1}{y-5} - 2} = \frac{11y}{11} = y.$$

Thus f is surjective. It follows that f is bijective. □

5.3. Composition of Functions

Let A, B and C be nonempty sets, and let $f : A \longrightarrow B$ and $g : B \longrightarrow C$ be two functions. We are going to create a new function from f and g, called their *composition*:

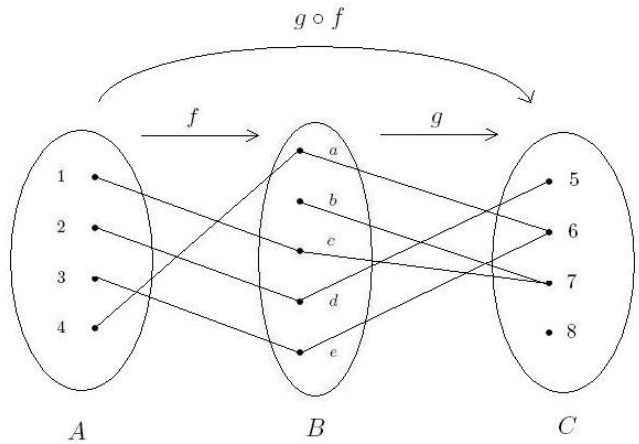
Definition 5.21 Let $f : A \longrightarrow B$ and $g : B \longrightarrow C$ be two functions. The *composition* of f and g, denoted by $g \circ f$, is the function from A to C defined by

$$(g \circ f)(a) = g(f(a)) \qquad \text{for all } a \in A.$$

The composition $g \circ f$ is obtained by first apply f and then apply g. The following is an example of composition of functions:

Example 5.22 Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c, d, e\}$ and $C = \{5, 6, 7, 8\}$. Define the functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$ by

$$f = \{(1, c),\ (2, d),\ (3, e),\ (4, a)\} \quad \text{and} \quad g = \{(a, 6),\ (b, 7),\ (c, 7),\ (d, 5),\ (e, 6)\}$$



According to the definition of $g \circ f$, we have $(g \circ f)(1) = g(f(1)) = g(c) = 7$. Other values are obtained similarly. Hence,

$$g \circ f = \{(1,7),\ (2,5),\ (3,6),\ (4,6)\}. \qquad \square$$

The fact that $g \circ f$ is defined does not necessarily imply that $f \circ g$ is also defined, since the domain of f may be different from the codomain of g. In the example above, $f \circ g$ is not defined. In order that both $g \circ f$ and $f \circ g$ are defined, it is necessary that $f : A \longrightarrow B$ and $g : B \longrightarrow A$. But then $g \circ f : A \longrightarrow A$ and $f \circ g : B \longrightarrow B$, and therefore $g \circ f \neq f \circ g$ unless $A = B$. Even if $A = B$, $g \circ f$ and $f \circ g$ may not be equal, as shown in the following example:

Example 5.23 Let $A = \{1, 2\}$ and let $f : A \longrightarrow A$, $g : A \longrightarrow A$ be defined by

$$f = \{(1,1),\ (2,1)\} \quad \text{and} \quad g = \{(1,2),\ (2,2)\}.$$

Then

$$g \circ f = \{(1,2),\ (2,2)\} = g \quad \text{and} \quad f \circ g = \{(1,1),\ (2,1)\} = f.$$

Since $f \neq g$ (recall Definition 5.8), we have $g \circ f \neq f \circ g$. $\square$

Remark: The composition of functions is in general not *commutative*.

Fortunately, the composition of functions is *associative*:

Theorem 5.24 *For any nonempty sets A, B, C, D, and for any functions $f : A \longrightarrow B$, $g : B \longrightarrow C$ and $h : C \longrightarrow D$, we have*

$$h \circ (g \circ f) = (h \circ g) \circ f.$$

Proof. First of all, $g \circ f$ is a function from A to C and $h \circ g$ is a function from B to D. Hence both $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are functions from A to D. It remains to verify that they map every $a \in A$ to the same image in D. Indeed,

$$(h \circ (g \circ f))(a) = h((g \circ f)(a)) = h(g(f(a)))$$

and

$$((h \circ g) \circ f)(a) = (h \circ g)(f(a)) = h(g(f(a))). \qquad \square$$

Example 5.25 Let A, B, C, f and g be given in Example 5.22. Let $D = \{x, y, z\}$ and define $h : C \longrightarrow D$ by $h = \{(5, y), (6, x), (7, z), (8, y)\}$. It is straightforward to see that both $h \circ (g \circ f)$ and $(h \circ g) \circ f$ are given by

$$h \circ (g \circ f) = \{(1, z), (2, y), (3, x), (4, x)\} = (h \circ g) \circ f. \qquad \square$$

Now we shall prove a theorem which tells us that the composition of two injective functions is injective. The same is also true for surjective functions:

Theorem 5.26 *For any nonempty sets A, B, C and for any functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$,*

(a) if f and g are injective, then so is $g \circ f$.

(b) if f and g are surjective, then so is $g \circ f$.

Proof. For (a), we need to show that for any $x, y \in A$,

$$(g \circ f)(x) = (g \circ f)(y) \implies x = y.$$

Assume that $(g \circ f)(x) = (g \circ f)(y)$. Then by the definition of composition, we have $g(f(x)) = g(f(y))$. But then we have $f(x) = f(y)$, since g is assumed to be injective. Now we can conclude that $x = y$, as f is also assumed to be injective.

For (b), we need to show that for any $c \in C$, there exists $a \in A$ such that $(g \circ f)(a) = c$. For any $c \in C$, since g is assumed to be surjective, there must exist $b \in B$ such that $g(b) = c$. Now for this $b \in B$, there exists $a \in A$ such that $f(a) = b$ as f is also assumed to be surjective. Hence $(g \circ f)(a) = g(f(a)) = g(b) = c$, implying that $g \circ f$ is surjective. $\square$

Combining the two results in Theorem 5.26 yields the following corollary:

Corollary 5.27 *For any nonempty sets A, B, C and for any functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$, if $f : A \longrightarrow B$ and $g : B \longrightarrow C$ are bijective, then so is $g \circ f$.*

It is interesting to know whether the converse of the above results hold. Details are left as exercises below:

Exercise For any nonempty sets A, B, C and for any functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$, prove each of the following statements:

(a) If $g \circ f$ is injective, then f is injective.

(b) If $g \circ f$ is surjective, then g is surjective.

(c) If $g \circ f$ is bijective, then f is injective and g is surjective. (This follows from (a) and (b).)

Exercise For any nonempty sets A, B, C and for any functions $f : A \longrightarrow B$ and $g : B \longrightarrow C$, prove or disprove each of the following statements:

(a) If $g \circ f$ is injective, then g is injective.

(b) If $g \circ f$ is surjective, then f is surjective.

(c) If $g \circ f$ is bijective, then f is surjective or g is injective.

5.4. Inverse Functions

Recall that a function $f : A \longrightarrow B$ is a relation from A to B satisfying the condition stated in Definition 5.1. We also recall from *Definition 4.7 (Chapter 4)* that given a relation R from A to B, one can always define the *inverse relation* R^{-1} from B to A by

$$R^{-1} = \{(b, a) : (a, b) \in R\}.$$

Then the following question arises naturally: Given a function $f : A \longrightarrow B$ which is a relation from A to B, under what conditions its inverse relation f^{-1} is also a function?

Let us begin with an example and see if this example shed some light on the answer to the above question. Let A, B and f be given in Example 5.3. Then the inverse relation f^{-1} is

$$f^{-1} = \{(a, 1), (a, 2), (c, 3), (c, 4), (c, 5), (e, 6)\}.$$

It is clear that f^{-1} does not define a function. The reasons the inverse relation f^{-1} fails to be a function are (i) there are elements in B (e.g. b) which are not related to any elements in A; (ii) there are elements in B (e.g. a) which are related to two or more elements in A.

If f^{-1} defines a function from B to A, then certainly we have $\operatorname{dom} f^{-1} = B$. This suggests that every element in B is related to some elements in A, that is, f has to be surjective. Furthermore, if

f^{-1} defines a function, each element b in B should have a unique image a in A. This suggests that f has to be injective, for if f is not injective, then there are distinct elements in A mapping into the same element in B, and we will have situation (ii) as above, which means that f^{-1} cannot define a function.

The above discussion shows that if f^{-1} defines a function from B to A, then f has to be injective and surjective, that is, f has to be bijective. The following theorem tells us that bijectivity is indeed both necessary and sufficient for the existence of the inverse function f^{-1}:

Theorem 5.28 *Let $f : A \longrightarrow B$ be a function. Then the inverse relation f^{-1} is a function from B to A if and only if f is bijective.*

Proof. First assume that f^{-1} is a function from B to A. We have to prove that f is injective and surjective.

Injective: For any $x, y \in A$, suppose that $f(x) = f(y) = z \in B$. Then (x, z), $(y, z) \in f$ and this means (z, x), $(z, y) \in f^{-1}$. Since f^{-1} is assumed to be a function, we must have $x = y$. Hence f is injective.

Surjective: For any $b \in B$, since f^{-1} is a function, there is a unique $a \in A$ such that $f^{-1}(b) = a$. Hence $(b, a) \in f^{-1}$ and so $(a, b) \in f$. It follows that we have $f(a) = b$. Hence f is surjective.

Next we assume that f is bijective. We need to show that f^{-1} is a function from B to A. For any $b \in B$, since f is surjective, there exists $a \in A$ such that $f(a) = b$. Thus we have $(a, b) \in f$ and hence $(b, a) \in f^{-1}$. It remains to show that such a is unique, that is, if (b, a') is also in f^{-1}, then $a = a'$.

If (b, a') is also in f^{-1}, then both (a, b), $(a', b) \in f$. This means $f(a) = f(a') = b$. Since f is assumed to be injective, we have $a = a'$ and we are done. $\square$

If $f : A \longrightarrow B$ is bijective, then by Theorem 5.28, f^{-1} is a function from B to A. In fact, f^{-1} is also bijective:

Theorem 5.29 *If $f : A \longrightarrow B$ is bijective, then $f^{-1} : B \longrightarrow A$ is also bijective.*

Proof. Theorem 5.28 tells us that if f is bijective, then f^{-1} is a function from B to A. First we show that f^{-1} is injective. For any $b, b' \in B$, suppose $f^{-1}(b) = f^{-1}(b') = a$, then (b, a), $(b', a) \in f^{-1}$ and hence (a, b), $(a, b') \in f$. Then we must have $b = b'$.

To show that f^{-1} is surjective, let $a \in A$. Since f is a function, there is a unique $b \in B$ such that $f(a) = b$. Therefore $(a, b) \in f$ and hence $(b, a) \in f^{-1}$. This means $f^{-1}(b) = a$, and so f^{-1} is surjective. $\square$

Let $f : A \longrightarrow B$ be a bijective function. By Theorem 5.29, $f^{-1} : B \longrightarrow A$ is a bijective function. We refer to this function as the *inverse function* of f. Then the composition function $f^{-1} \circ f$ makes sense, and it is a function from A to A. Similarly, $f \circ f^{-1}$ also makes sense, and is a function from B to B. Now it is not hard to see that

$$f^{-1} \circ f = id_A \quad \text{and} \quad f \circ f^{-1} = id_B$$

where id_A and id_B are respectively the identity functions (Example 5.7) on A and B. We finish this section by studying an approach which makes use of the above composition functions to finding a formula for the inverse function:

Example 5.30 The function $f : \mathbb{R} \setminus \{2\} \longrightarrow \mathbb{R} \setminus \{5\}$ defined by $f(x) = \dfrac{5x+1}{x-2}$ is known to be bijective from Example 5.20. Determine $f^{-1}(x)$ for $x \in \mathbb{R} \setminus \{5\}$.

Solution. Since $(f \circ f^{-1})(x) = x$ for all $x \in \mathbb{R} \setminus \{5\}$, we have

$$x = (f \circ f^{-1})(x) = f(f^{-1}(x)) = \frac{5f^{-1}(x)+1}{f^{-1}(x)-2}.$$

Then we have $xf^{-1}(x) - 2x = 5f^{-1}(x) + 1$, and hence

$$f^{-1}(x) = \frac{2x+1}{x-5} \quad \text{for } x \in \mathbb{R} \setminus \{5\}.$$

□

Remark: Note that this method does not work for functions such as $f(x) = x^7 + x^3 + 1$, $\sin x$, $e^x, \ldots$.

5.5. Image Sets and Preimage Sets

Definition 5.31 Let A and B be sets and let $f : A \longrightarrow B$ be a function.

(a) Suppose $X \subseteq A$. The *image (set)* of X is the set

$$f(X) = \{y \in B : y = f(x) \text{ for some } x \in X\} \subseteq B.$$

(b) Suppose $Y \subseteq B$. The *preimage (set)* of Y is the set

$$f^{-1}(Y) = \{x \in A : f(x) \in Y\} \subseteq A.$$

Remarks

(i) It is important to realize that X and Y in Definition 5.31 are subsets, but not elements, of A and B respectively.

(ii) Note that the preimage $f^{-1}(Y)$ of Y still makes sense even if f is not bijective. The expression $f^{-1}(Y)$ stands for the set of elements in A that f sends into Y.

Example 5.32 Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{a, b, c, d, e\}$. Let f be the function given in Example 5.3:

$$f = \{(1, a), (2, a), (3, c), (4, c), (5, c), (6, e)\}.$$

For example, we have

(i) $f(\{1, 6\}) = \{a, e\}$.

(ii) $f(\{3, 4, 5\}) = \{c\}$.

(iii) $f(\{1, 2, 3, 4, 5, 6\}) = \{a, c, e\}$.

(iv) $f^{-1}(\{a\}) = \{1, 2\}$.

(v) $f^{-1}(\{b\}) = \emptyset$.

(vi) $f^{-1}(\{c, d, e\}) = \{3, 4, 5, 6\}$.

Note that f is not bijective, but it is meaningful to consider the preimage sets of f. For example, we have $f^{-1}(\{a\}) = \{1, 2\}$, though $f^{-1}(a)$ has absolutely no meaning because the inverse function f^{-1} does not exist. □

Exercise Consider the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by $f(x) = x^2$.

(i) Find $f(\{4\})$ and $f([-1, 3])$.

(ii) Find $f^{-1}(\{9\})$, $f^{-1}(\{-4\})$ and $f^{-1}([-1, 9])$.

Observe that in (i) of Example 5.32, we have $f(\{1, 6\}) = \{a, e\}$. But $f^{-1}(\{a, e\}) = \{1, 2, 6\} \neq \{1, 6\}$. This suggests that $f^{-1}(f(X)) \neq X$ in general. Similarly, we have $f^{-1}(\{c, d, e\}) = \{3, 4, 5, 6\}$, but $f(\{3, 4, 5, 6\}) = \{c, e\}$. This suggests that $f(f^{-1}(Y)) \neq Y$ in general. However, we have the following result:

Theorem 5.33 *Let A and B be sets and let $f : A \longrightarrow B$ be a function. Then*

(a) $X \subseteq f^{-1}(f(X))$ for any subset $X \subseteq A$ and

(b) $f(f^{-1}(Y)) \subseteq Y$ for any subset $Y \subseteq B$.

Proof.

(a) For any subset $X \subseteq A$, we have $f^{-1}(f(X)) = \{x \in A : f(x) \in f(X)\}$ by putting $Y = f(X)$ in the definition of preimage sets. Now take any $x \in X$, then $f(x) \in f(X)$ and hence $x \in f^{-1}(f(X))$.

(b) For any subset $Y \subseteq B$, we have $f(f^{-1}(Y)) = \{y \in B : y = f(x)$ for some $x \in f^{-1}(Y)\}$ by putting $X = f^{-1}(Y)$ in the definition of image sets. Now take any $y \in f(f^{-1}(Y))$, then $y = f(x)$ for some $x \in f^{-1}(Y)$. But $x \in f^{-1}(Y)$ means $f(x) \in Y$. Hence $y = f(x) \in Y$. □

5.6. Cardinalities of Sets

We have studied the cardinality $|A|$ of a finite set A, which is just the number of elements in A. In this section we are going to study the cardinality of infinite sets, and in particular when we can say that two infinite sets are of the same cardinality. The key idea here is to make use of the concept of bijective functions.

Let us begin with a simple example. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{2, 4, 6, 8, 10\}$. Both A and B are finite sets with $|A| = |B| = 5$. We can also compare the two sets by the following way:

$$\begin{array}{cccccc} A & 1 & 2 & 3 & 4 & 5 \\ & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ B & 2 & 4 & 6 & 8 & 10 \end{array}$$

Observe that we can pair off the elements of A and B. We say that there is a *one-to-one correspondence* between elements of the two sets. More precisely, the above way of pairing off the elements defines a bijective function $f : A \longrightarrow B$ given by

$$f = \{(1, 2), (2, 4), (3, 6), (4, 8), (5, 10)\}.$$

This leads us to the following definition:

Definition 5.34 Two sets A and B are said to have the *same cardinality*, written $|A| = |B|$, if either A and B are both empty or there is a bijective function f from A to B. Two sets having the same cardinality are also refered to as *numerically equivalent sets*.

We can now apply Definition 5.34 to compare the "size" of infinite sets:

Example 5.35 Prove that the set of integers $\mathbb{Z}$ and the set of even integers $E = \{2x : x \in \mathbb{Z}\}$ are numerically equivalent.

Proof. Define the function $f : \mathbb{Z} \longrightarrow E$ by $f(x) = 2x$. Then f is a bijection (check!) between $\mathbb{Z}$ and E, and hence $\mathbb{Z}$ and E are numerically equivalent. □

One may find the result in Example 5.35 unsettling. The set $\mathbb{Z}$ of integers seems twice as large as the set of even integers as $\mathbb{Z}$ contains both the odd integers as well as the even integers. The set of even integers is certainly a proper subset of $\mathbb{Z}$, but it turns out that the two sets are of the same cardinality. Such kind of phenomenon is common when dealing with infinite sets.

Exercise Prove that the set of integers $\mathbb{Z}$ and the set of odd integers are numerically equivalent.

Exercise For any sets A, B and C, prove each of the following statements:

(i) A is numerically equivalent to A.

(ii) If A is numerically equivalent to B, then B is numerically equivalent to A.

(iii) If A is numerically equivalent to B, and B is numerically equivalent to C, then A is numerically equivalent to C.

Now let us focus on the set $\mathbb{N}$ of natural numbers and consider those sets which have the same cardinality as $\mathbb{N}$:

Definition 5.36 A set A is called *denumerable* if A is numerically equivalent to $\mathbb{N}$, that is, there exists a bijective function $f : \mathbb{N} \to A$.

Remarks

(i) Certainly, $\mathbb{N}$ itself is denumerable.

(ii) If A is a denumerable set, then A is an infinite set. Moreover, the bijective function $f : \mathbb{N} \to A$ allows us to list out the set A as an infinite list $f(1), f(2), f(3), \ldots$. Conversely, if the elements of A can be listed as $a_1, a_2, a_3, \ldots$ with $a_i \neq a_j$ for $i \neq j$, then we can define a function $f : \mathbb{N} \to A$ by $f(n) = a_n$. This function is then a bijection, and hence A is denumerable.

(iii) The cardinality of $\mathbb{N}$ is denoted by $|\mathbb{N}| = \aleph_0$. Thus any denumerable set has cardinality $\aleph_0$.

Definition 5.37 A set A is *countable* if it is either finite or denumerable. A set is not countable is called *uncountable*.

Remark Countably infinite sets are exactly the denumerable sets. An uncountable set is necessarily infinite.

Example 5.38 Prove that the set $\mathbb{Z}$ of integers is denumerable.

Proof. This means that we need to construct a bijective function $f : \mathbb{N} \longrightarrow \mathbb{Z}$. The following suggests a way of writing down such a function:

$$\begin{array}{cccccccccc}
\mathbb{N} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & \ldots \\
 & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \\
\mathbb{Z} & 0 & 1 & -1 & 2 & -2 & 3 & -3 & \ldots
\end{array}$$

Observe that given any $x \in \mathbb{Z}$ there is exactly one natural number $n \in \mathbb{N}$ such that $f(n) = x$, so f is bijective. Therefore $\mathbb{Z}$ is denumerable. $\square$

Remarks

(i) The function $f : \mathbb{N} \longrightarrow \mathbb{Z}$ above can also be defined by the formula

$$f(n) = \frac{1 + (-1)^n(2n-1)}{4}.$$

(ii) Again, it is possible for two sets to have the same cardinality where one is a proper subset of the other. Here we have $\mathbb{N} \subset \mathbb{Z}$ and $|\mathbb{N}| = |\mathbb{Z}|$.

(iii) Example 5.35 and the exercise following it suggest that both the set of even and odd integers are denumerable.

Now we know that $\mathbb{N}$ and $\mathbb{Z}$ have the same cardinality. What about the set $\mathbb{Q}$ of rational numbers? Is $\mathbb{Q}$ denumerable or uncountable? Since $\mathbb{Z} \subset \mathbb{Q}$, we expect that the set $\mathbb{Q}$ is much larger than $\mathbb{Z}$. However it is a surprising fact that $\mathbb{Q}$ is denumerable. We first show that the set $\mathbb{Q}^+$ of positive rational numbers is denumerable:

Theorem 5.39 *The set $\mathbb{Q}^+$ of positive rational numbers is denumerable.*

Proof. To prove this we need to show how to write down all the rational numbers p/q in list form. This is done by arranging these numbers in an infinite array:

	1	2	3	4	5	...
1	$\frac{1}{1}$	$\frac{2}{1}$	$\frac{3}{1}$	$\frac{4}{1}$	$\frac{5}{1}$	...
2	$\frac{1}{2}$	$\frac{3}{2}$	$\frac{5}{2}$	$\frac{7}{2}$	$\frac{9}{2}$	...
3	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{4}{3}$	$\frac{5}{3}$	$\frac{7}{3}$	...
4	$\frac{1}{4}$	$\frac{3}{4}$	$\frac{5}{4}$	$\frac{7}{4}$	$\frac{9}{4}$	...
5	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{3}{5}$	$\frac{4}{5}$	$\frac{6}{5}$	...
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	

In the first row, all positive rational numbers j/i with $i = 1$ are shown. In the second row, all positive rational numbers j/i with $i = 2$ and such that j/i has been reduced to lowest terms are shown. We continue in this manner with all other rows. In this way, every positive rational number appears exactly once in the array. Beginning at $1/1$ and following the arrows, we get an infinite list of all rational numbers:

$$\frac{1}{1}, \frac{2}{1}, \frac{1}{2}, \frac{3}{1}, \frac{3}{2}, \frac{1}{3}, \frac{4}{1}, \frac{5}{2}, \frac{2}{3}, \frac{1}{4}, \ldots$$

We can now describe a bijective function $f : \mathbb{N} \longrightarrow \mathbb{Q}^+$ according to the above list:

$$f(1) = \frac{1}{1} = 1,\ f(2) = \frac{2}{1} = 2,\ f(3) = \frac{1}{2},\ f(4) = \frac{3}{1} = 3,\ f(5) = \frac{3}{2}, \ldots$$

It follows that $\mathbb{Q}^+$ is denumerable. □

Remark The function described above is by no means unique. There are other ways to snake around the array so that every number is being visited. However, traversing the numbers by rows does not work. If we start from the first row, we will only encounter the positive integers as the first row never ends.

We have shown that $\mathbb{Q}^+$ is denumerable, it is then not hard to show that $\mathbb{Q}$ is also denumerable:

Theorem 5.40 *The set $\mathbb{Q}$ of all rational numbers is denumerable.*

Proof. Since $\mathbb{Q}^+$ is denumerable, we can write $\mathbb{Q}^+ = \{q_1, q_2, q_3, \dots\}$. Thus

$$\mathbb{Q} = \{q_1, q_2, q_3, \dots\} \cup \{0\} \cup \{-q_1, -q_2, -q_3, \dots\} = \{0, q_1, -q_1, q_2, -q_2, q_3, -q_3, \dots\}.$$

Then the function $f : \mathbb{N} \longrightarrow \mathbb{Q}$ defined by

$$f(1) = 0,\ f(2) = q_1,\ f(3) = -q_1,\ f(4) = q_2,\ f(5) = -q_2, \dots$$

is bijective and hence $\mathbb{Q}$ is denumerable. □

The examples we have seen so far are denumerable sets, in particular we have $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}|$. For uncountable sets, the best known example is the set $\mathbb{R}$ of all real numbers. This means that there does not exist a bijective function $f : \mathbb{N} \longrightarrow \mathbb{R}$. The uncountability of $\mathbb{R}$ suggests that the "size" of it is different from the "size" of $\mathbb{N}$ – there are different sizes of infinity!

The proof that $\mathbb{R}$ is uncountable was established by G. Cantor who applied a *diagonal argument* which can also be used to show that several other sets are uncountable. We will not study the details of the argument here, but the following illustrates the argument on the open interval $(0, 1)$, and show that there does not exist a bijective function $f : \mathbb{N} \longrightarrow (0, 1)$:

Theorem 5.41 *The open interval $(0, 1)$ is uncountable.*

Proof. Omitted. (See, for instance, Theorem 10.8 of *Mathematical Proofs: A Transition to Advanced Mathematics, 2nd edition*) □

Theorem 5.42 *The open interval $(0, 1)$ and $\mathbb{R}$ are numerically equivalent.*

Proof. Omitted. (See, for instance, Theorem 10.13 of *Mathematical Proofs: A Transition to Advanced Mathematics, 2nd edition*)

Theorem 5.41 and 5.42 imply that $\mathbb{R}$ is uncountable:

Theorem 5.43 *The set $\mathbb{R}$ of real numbers is uncountable.*

We state (without proof) here several results that may be useful for comparing denumerable or uncountable sets:

Theorem 5.44 *Every infinite subset of a denumerable set is denumerable.*

Proof. Omitted. (See, for instance, Theorem 10.3 of *Mathematical Proofs: A Transition to Advanced Mathematics, 2nd edition*) □

Theorem 5.45 *Let A and B be sets such that $A \subseteq B$. If A is uncountable, then B is uncountable.*

Proof. Omitted. (See, for instance, Theorem 10.9 of *Mathematical Proofs: A Transition to Advanced Mathematics, 2nd edition*) □

Remark We may also apply Theorem 5.41 and 5.45 to give an alternate proof that $\mathbb{R}$ is uncountable, as $(0, 1)$ is uncountable and $(0, 1) \subseteq \mathbb{R}$.

Theorem 5.46 *If A and B are denumerable sets, then $A \times B$ is denumerable.*

Proof. Omitted. (See, for instance, Result 10.5 of *Mathematical Proofs: A Transition to Advanced Mathematics, 2nd edition*) □

Theorem 5.47 *If A and B are denumerable sets, then $A \cup B$ is denumerable.*

Proof. Omitted. (c.f. Exercise 10.1 of *Mathematical Proofs: A Transition to Advanced Mathematics, 2nd edition*) □

Exercise

(a) Prove or disprove:

  (i) $\mathbb{N} \times \mathbb{N}$ is denumerable.

  (ii) $\mathbb{Z} \times \mathbb{Q}$ is denumerable.

  (iii) The closed interval $[0, 1]$ is uncountable.

  (iv) There exists a bijective function $f : \mathbb{Q} \longrightarrow \mathbb{R}$.

(b) Let $S \subseteq \mathbb{Z} \times \mathbb{Z}$ be defined by $S = \{(2x, 3x) : x \in \mathbb{Z}\}$. Prove that S is denumerable.

(c) Prove that for any finite subset S of $\mathbb{N}$, $|\mathbb{N} \setminus S| = |\mathbb{N}|$.

Comparing Cardinalities of Sets

So far we have been considering the situation when two sets A and B have the same cardinality. In this section we would like to extend the idea of equality between cardinalities by introducing the concept of "smaller" (and "larger") cardinality.

Definition 5.48 Let A and B be sets.

(a) $|A| \leq |B|$ means there is an injective function $f : A \longrightarrow B$.

(b) $|A| < |B|$ means $|A| \leq |B|$ and $|A| \neq |B|$, that is, there is an injective function $f : A \longrightarrow B$, but NO bijective function from A to B.

Remarks

(i) If the two sets are finite, we certainly know what "smaller" means – the set A contains fewer elements than B. The above definition not only gives exactly this idea when A and B are finite, but it also includes the situation for infinite sets.

(ii) The existence of an injective function from A to B tells us that we can pair off the elements of A with some (or all) of the elements of B. In the case $|A| < |B|$, the we cannot pair off the elements of A with all of the elements of B.

(iii) The concept of "larger" cardinality is as follows: $|A| \geq |B|$ means $|B| \leq |A|$; and $|A| > |B|$ means $|B| < |A|$.

Exercise

(a) Prove that for any sets A and B, if $A \subseteq B$, then $|A| \leq |B|$.
(This result says in particular that $|A| \leq |A|$ for any set A.)

(b) Prove that for any sets A, B and C, if $A \subseteq B$ and $B \subseteq C$, then $|A| \leq |C|$.

It follows from Exercise (a) that $|\mathbb{N}| \leq |\mathbb{R}|$. Moreover, Cantor's diagonal argument suggests that there is no bijective function from $\mathbb{N}$ to $\mathbb{R}$. Consequently, we have

$$|\mathbb{N}| < |\mathbb{R}|.$$

Two natural questions then arise – (1) is there a set S for which $|\mathbb{N}| < |S| < |\mathbb{R}|$? (2) is there a set S for which $|\mathbb{R}| < |S|$? The answer to (2) is "yes", and we will explain its solution after the discussion on (1).

Many mathematicians struggled with question (1) – the existence of a set S with "intermediate size". The suggestion that no such S exists is known as the *Continuum Hypothesis*. In other words, the Continuum Hypothesis asserts that no set has a cardinality between that of $\mathbb{N}$ and $\mathbb{R}$. This interesting conjecture was posed by Cantor, and became a famous unresolved issue for many years. A remarkable result of Gödel and Cohen showed that the Continuum Hypothesis is *undecidable*, meaning that it cannot be proved nor disproved from the axioms on which set theory is based. The hypothesis is therefore *independent* of the axioms of set theory.

For question (2) the following theorem will explain its solution:

Theorem 5.49 *For any set A, we have $|A| < |\mathcal{P}(A)|$.*

Proof. First of all, we remark that this theorem is obvious when A is an empty set or a finite set. If A has n elements for some $n \in \mathbb{N} \cup \{0\}$, then $\mathcal{P}(A)$ has 2^n elements (see *Example 3.37 (Chapter 3)*) and we have $n < 2^n$. However, the proof given here applies to *any* sets A, both finite and infinite.

Let A be any set. We may assume that A is nonempty as the theorem is true for the case $A = \emptyset$. According to Definition 5.48 (b), we need to show that $|A| \leq |\mathcal{P}(A)|$ and $|A| \neq |\mathcal{P}(A)|$. To show $|A| \leq |\mathcal{P}(A)|$, we have to find an injective function $f : A \longrightarrow \mathcal{P}(A)$. Indeed, it is not hard to construct such a function – define $f : A \longrightarrow \mathcal{P}(A)$ by $f(a) = \{a\}$ for any $a \in A$. Then f is clearly injective since if $f(a) = f(b)$, then $\{a\} = \{b\}$ and hence $a = b$.

Next we will show that $|A| \neq |\mathcal{P}(A)|$. Assume to the contrary that $|A| = |\mathcal{P}(A)|$. Then there exists a bijective function $g : A \longrightarrow \mathcal{P}(A)$. We will try to obtain a contradiction (to the fact that g is bijective) by exhibiting a subset X of A such that $X \neq g(a)$ for any $a \in A$.

For any $a \in A$, $g(a) \in \mathcal{P}(A)$ is a subset of A. Hence there are two cases: either (i) $a \in g(a)$, or (ii) $a \notin g(a)$. Then we define a subset X by

$$X = \{a \in A : a \notin g(a)\}.$$

If $a \in g(a)$ (case (i)), then $a \notin X$ by the definition of X. This implies that $g(a) \neq X$. On the other hand, if $a \notin g(a)$ (case (ii)), then $a \in X$ by the definition of X. We again have the same conclusion that $g(a) \neq X$. Consequently, we have $g(a) \neq X$ for any $a \in A$. This contradicts to the fact that g is bijective, and therefore we must have $|A| \neq |\mathcal{P}(A)|$. □

Remarks

(a) According to Theorem 5.49, there is no "largest" set – if we begin with the set $\mathbb{R}$, we have the following chain of infinite cardinalities:

$$|\mathbb{R}| < |\mathcal{P}(\mathbb{R})| < |\mathcal{P}(\mathcal{P}(\mathbb{R}))| < \cdots$$

(b) The answer to question (2) is now clear – there does exist a set S, say $S = \mathcal{P}(\mathbb{R})$, such that $|\mathbb{R}| < |S|$.

We will end this section by stating the following famous result which provides an alternative way for proving that two sets have the same cardinality:

Theorem 5.50 (Schröder-Bernstein Theorem) *For any sets A and B, if $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.*

This result is also referred to as the *Cantor-Schröder-Bernstein Theorem*.

In terms of Definition 5.48, this result states that if there exists an injective function $f : A \longrightarrow B$, and an injective function $g : B \longrightarrow A$, then there exists a bijective function $h : A \longrightarrow B$. The theorem is trivial in the case of finite sets, but is highly nontrivial in the case of infinite sets.

This result is useful in the sense that finding two injective functions – one from A to B and the other from B to A – is often far easier than finding a single bijective function. Since the proof of this theorem is rather subtle, we shall skip it (see, for instance, Section 10.5 of *Mathematical Proofs: A Transition to Advanced Mathematics, 2nd edition*) and discuss here some of its applications:

Example 5.51 Use the Schröder-Bernstein Theorem to prove that $|(0,1)| = |[0,1]|$.

Proof. To apply the Schröder-Bernstein Theorem, we shall need to construct an injective function $f : (0,1) \longrightarrow [0,1]$ and an injective function $g : [0,1] \longrightarrow (0,1)$. Since $(0,1) \subseteq [0,1]$, we can simply define $f : (0,1) \longrightarrow [0,1]$ by $f(x) = x$, which is clearly injective.

Next we define the function $g : [0,1] \longrightarrow (0,1)$ by $g(x) = \frac{x+1}{3}$. One can imagine that this function g shrinks the interval $[0,1]$ to $[\frac{1}{3}, \frac{2}{3}] \subseteq (0,1)$. It is easy to see that g is injective and hence, by the Schröder-Bernstein Theorem, there exists a bijective function from $(0,1)$ to $[0,1]$, that is, we have $|(0,1)| = |[0,1]|$. □

Exercise

(a) Use the Schröder-Bernstein Theorem to prove that $|(0,1)| = |[0,1)| = |(0,1]|$.

(b) Use the Schröder-Bernstein Theorem to prove that $|[1,3]| = |(5,8)|$.

Example 5.52 Use the Schröder-Bernstein Theorem to prove that $|\mathbb{R}| = |\mathbb{R} \setminus \{0\}|$.

Proof. The function $f : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}$ given by $f(x) = x$ is injective. Moreover, we define $g : \mathbb{R} \longrightarrow \mathbb{R} \setminus \{0\}$ by

$$g(x) = \begin{cases} x & \text{if } x > 0, \\ x - 1 & \text{if } x \leq 0. \end{cases}$$

Then g is also injective and so by the Schröder-Bernstein Theorem, we have $|\mathbb{R}| = |\mathbb{R} \setminus \{0\}|$. □